

JIANYU (JIM) AN

jianyu34@umd.edu

EDUCATION

- PhD, Computer Science, **University of Maryland**, 2028 (projected)
 - GAMMA group with Prof Dinesh Manocha
 - Relevant Coursework: Decision Making for Robotics, Long Context Language Models, Generative AI Agents
 - GPA 3.85
- MS, Computer Science, **Georgia Institute of Technology**, 2023
 - Relevant Coursework: Artificial Intelligence, Deep Learning, Reinforcement Learning, High Performance Computing
 - GPA 3.60
- BS, Engineering Science and Mechanics, **Virginia Polytechnic Institute and State University**, 2016
 - Minors in Mathematics, Biomedical Engineering,
 - GPA: 3.82

EXPERIENCE

Research Intern, Skyline Nav AI

Summer 2025

- Experimented with various state of the art vision based localization methods for UAVs in GPS denied environments

AI Researcher, Autonomy and AI Applications Division, NAVAIR

Sept 2019 to Dec 2024

- Gaps Alignment Project Tool (GAPT) project
 - Generated keywords and document similarities using natural language processing
- Confrontation of Adversarial Neural Networks (CANN) project
 - Developed competing machine learning agents in a unity-based simulation environment
 - Aim to identify novel strategies and tactics from agent interactions
- Machine Learning for Combat ID project
 - Team lead, Constructed a neural network architecture to classify communications modulations
- RAIDER CODE Autonomy team
 - Participated in developing new autonomy behavior and test scenarios
 - Conducted flight experiments demonstrating collaborative behavior in real and virtual assets

Operations Research Analyst, Mission Engineering and Analysis Division, NAVAIR

Aug 2016 to Jan 2020

- Worked with Campaign(STORM) and Mission(AFSIM) Level models to conduct trade studies
- Used Design of Experiments (DOE) to conduct a study on the impact of readiness levels of F-18s

Flight Test Engineer

- Facilitated test cards approval, Participated in flight tests, and Authored test transcripts

PRESENTATIONS AND PUBLICATIONS

- Sakano, K., **An, J.**, Mockler, J., Xu, H., Manocha, D. (2025). SAFE-SMART: Safety Analysis and Formal Evaluation using STL Metrics for Autonomous Robots. Under Review.
- Seneviratne, G., **An, J.**, Ellahy, S., Weerakoon, K., Elnoor, M.B., Kannan, J.D., Sunil, A.T., Manocha, D. (2025). HALO: Human Preference Aligned Offline Reward Learning for Robot Navigation. Conference on Robot Learning (CoRL), Seoul, Korea.
- **An, J.**, Ortiz, A., (2022). AI Training Framework for a Medium Fidelity Environment. Naval Applications of Machine Learning (NAML), San Diego, CA.
- **An, J.**, Kocan, K. (2019). Machine Learning Application for Combat Identification. NAWCAD Young Professional Symposium, Patuxent River, MD.
- Kim, S. J., Fezzaa, K., **An, J.**, Sun, T., Jung, S. (2017). Capillary spreading of contact line over a sinking sphere. Applied Physics Letters. 111. 134102. 10.1063/1.4991361.

OTHER EXPERIENCE

- 2016 - 2020 NAVAIR Professional Developmental Council
 - Led monthly meetings for developmental employees
 - Introduced new hires to the department, Mentored new employees
- 2016 - 2020 Emergency Medical Technician, NREMT-B
 - Provider in Charge, Lexington Park, MD
- 2013 - 2016 Virginia Tech: Various Research Labs
 - CRASH Lab: Analyzed the locomotion of a stingray and constructed a robotic prototype
 - Bio-Inspired Fluids Lab: Analyzed high-speed video of sinking spheres over various liquids
 - Mechanics of Soft Biological Systems Lab: Explored bi-axial mechanical properties of biological tissue

SKILLS

- Python, Pytorch, Tensorflow/Keras, Machine Learning, Reinforcement Learning, Modeling and Simulation